

Two Dimensional CNN Replication Report

NSL KDD and UNSW NB15

Result. The reported 2D CNN accuracy was reproduced for both datasets at the precision published in the paper. The experiment did not reproduce every loss and per-class value simultaneously.

Reference. Sharma et al., Explainable artificial intelligence for intrusion detection in IoT networks, Expert Systems with Applications 238 (2024) 121751.

Scope. The accuracy reproduction follows the paper's 2D CNN settings. The paper's LIME and SHAP examples were produced for its DNN; this report applies the same example-class protocol to our reproduced 2D CNN checkpoints.

Model Configuration

Setting	Value
Convolution filters	64, 32, 32
Kernel and activation	3 x 3, ReLU
Pooling	2 x 2 max pooling after each convolution
Output	5 neurons, softmax
Optimization	learning rate 0.001; weight decay 0.0001
Training	20 epochs; sparse categorical cross entropy
Data split	60% training; 15% validation; 25% testing

Dataset Preparation

Dataset	Records used	Classes	CNN input	Preparation
NSL-KDD	125,973	DoS, Normal, Probe, R2L, U2R	6 x 6 x 1	Label encoding, min-max scaling, six correlated columns removed
UNSW-NB15	185,124	DoS, Exploits, Fuzzers, Generic, Normal	7 x 7 x 1	50,000 Normal and Generic samples; 38 predictors plus 11 zeros

Concurrent Experiments

Dataset	Configurations	Seeds	Models at once	Runtime
NSL-KDD	12	64	768	50 min 18 s
UNSW-NB15	6	64	384	44 min 04 s

Grouped CUDA convolutions kept every model independent while executing all replicas in one process. Gradient accumulation preserved batch sizes 32, 64, and 128. AdamW and coupled-L2 Adam covered the paper's unspecified decay semantics.

Accuracy Replication

Dataset	Paper	Closest run	Configuration and seed	Absolute error	Match
NSL-KDD	0.994	0.993998857	batch032-zero_pad-adamw / 827550575	0.000001143	Yes
UNSW-NB15	0.81	0.810004322	batch064-adamw / 888329628	0.000004322	Yes

Closest Full Metric Profiles

Dataset	Configuration and seed	Accuracy	Loss	Raw RMSE	Rounded matches
NSL-KDD	batch032-zero_pad-adam_l2 / 1414324351	0.992030	0.026190	0.016176	11 / 17
UNSW-NB15	batch032-adamw / 1349501436	0.814326	0.395752	0.010835	7 / 17

Per Class Results for Closest Profiles

NSL KDD

Class	Paper P	Run P	Paper R	Run R	Paper F1	Run F1
DoS	1.00	0.998	1.00	0.997	1.00	0.998
Normal	0.99	0.989	1.00	0.996	0.99	0.993
Probe	0.98	0.989	0.99	0.986	0.99	0.987
R2L	0.93	0.939	0.51	0.558	0.66	0.700
U2R	0.60	0.600	0.23	0.231	0.33	0.333

UNSW NB15

Class	Paper P	Run P	Paper R	Run R	Paper F1	Run F1
DoS	0.57	0.566	0.04	0.038	0.07	0.071
Exploits	0.69	0.686	0.91	0.920	0.78	0.786
Fuzzers	0.60	0.621	0.80	0.781	0.69	0.692
Generic	1.00	0.992	0.97	0.978	0.99	0.985
Normal	0.91	0.929	0.81	0.827	0.86	0.875

LIME and SHAP Results

The closest individual 17-metric run for each dataset was extracted from the final concurrent checkpoint. Single-model inference agreed with the recorded grouped-model accuracy within the stated 0.0005 tolerance.

Dataset	Configuration and seed	Stored accuracy	Verified accuracy	Difference
NSL-KDD	batch032-zero_pad-adam_l2 / 1414324351	0.992030228	0.992157236	0.000127008
UNSW-NB15	batch032-adamw / 1349501436	0.814325843	0.814174589	0.000151253

Explanation Settings

Method	Setting	Value
LIME	Local perturbations	5,000 binary keep-or-training-mean samples
LIME	Surrogate	Exponential locality kernel and weighted ridge regression
SHAP	Local explanation	4,096 Kernel SHAP coalitions
SHAP	Global importance	20 stratified test cases; 768 coalitions per case
Both	Baseline	Training-feature mean
Both	Padding	Constant zero-padding excluded from explanations and fixed in model input

Local Explanation Checks

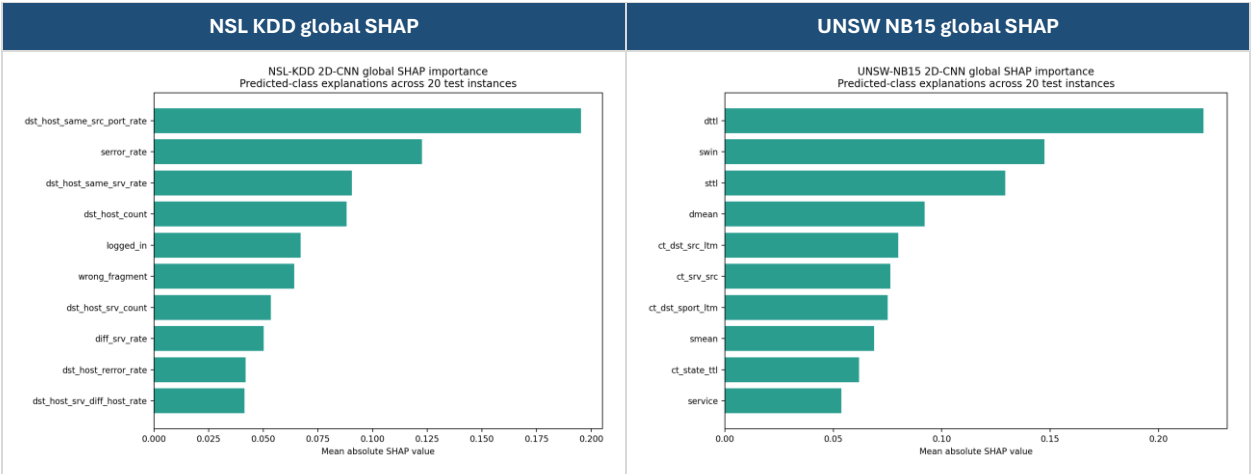
Dataset	LIME class	Probability	Weighted R2	Weighted RMSE
NSL-KDD	Normal	1.0000	0.2712	0.0080
NSL-KDD	DoS	0.9999	0.8701	0.0963
UNSW-NB15	Normal	1.0000	0.0418	0.0280
UNSW-NB15	Exploits	0.9182	0.9417	0.0770

Dataset	SHAP class	Base value	Prediction	Efficiency error
NSL-KDD	DoS	0.0011	0.9999	1.11e-16
UNSW-NB15	Normal	0.0032	1.0000	0.00e+00

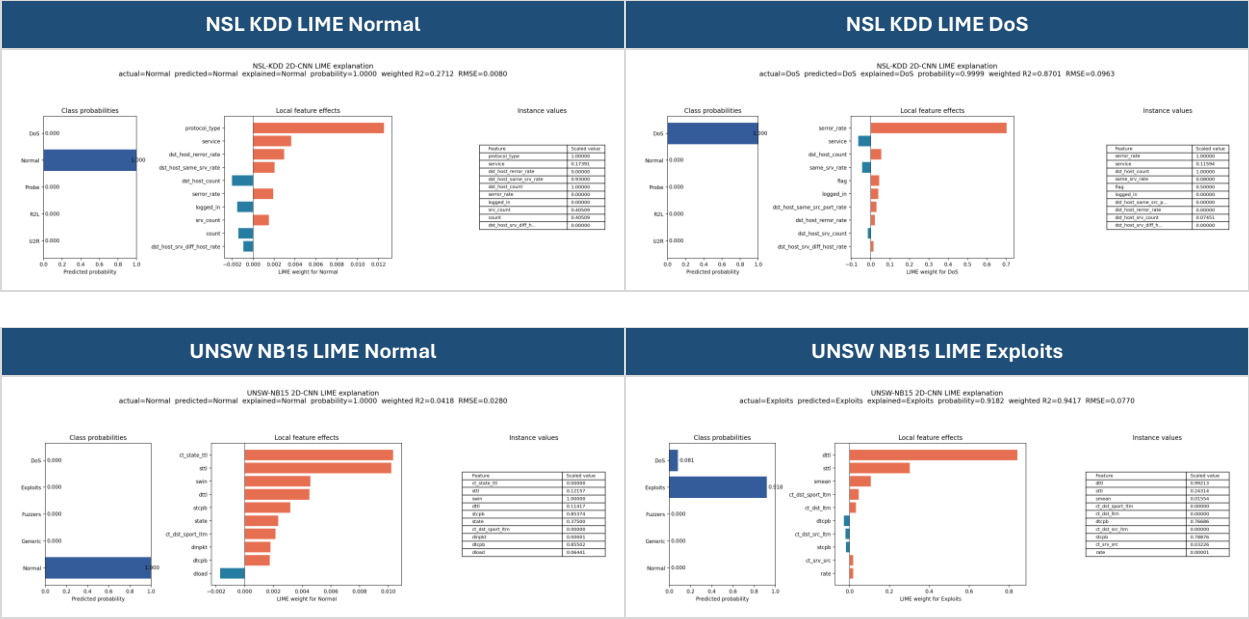
Interpretation. The low LIME R2 values for saturated Normal predictions show that a local linear surrogate explains limited probability variation around those instances. The small weighted RMSE values are reported beside R2 so this limitation is visible.

Global SHAP Importance

Rank	NSL KDD feature	Mean absolute SHAP	UNSW NB15 feature	Mean absolute SHAP
1	dst_host_same_src_port_rate	0.195277	dttl	0.220651
2	seerror_rate	0.122653	swin	0.147423
3	dst_host_same_srv_rate	0.090609	sttl	0.129387
4	dst_host_count	0.088175	dmean	0.092085
5	logged_in	0.067167	ct_dst_src_ltm	0.079936



Paper Class Explanation Examples



Code

Dataset	Preprocessing and model	Concurrent GPU training	Paper comparison
NSL-KDD	ns1_kdd_2dcnn.py	ns1_kdd_gpu_all_concurrent.py	compare_results_to_paper.py
UNSW-NB15	unsw_nb15_2dcnn.py	unsw_nb15_gpu_all_concurrent.py	compare_unsw_results_to_paper.py

Shared code	Purpose
explain_2dcnn.py	Checkpoint extraction, verified inference, LIME, local SHAP, global SHAP, plots, CSV, and JSON
test_explain_2dcnn.py	Five tests for reconstruction, surrogate behavior, padding exclusion, and regression calculations

```
# NSL-KDD
python .\ns1_kdd_gpu_all_concurrent.py
python .\compare_results_to_paper.py

# UNSW-NB15
python .\unsw_nb15_gpu_all_concurrent.py
python .\compare_unsw_results_to_paper.py

# Explain both selected 2D CNN runs
python .\explain_2dcnn.py --dataset both
python -m unittest -v .\test_explain_2dcnn.py
```

Conclusion. Both datasets matched the paper's reported 2D CNN accuracy after applying the stated preprocessing, architecture, split, and training settings. LIME and Kernel SHAP were then reproduced for the selected 2D CNN checkpoints with auditable sampling settings, verification tolerances, plots, CSV files, and JSON summaries.